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In [5]: import numpy as np
        from scipy.optimize import fsolve

        # Antoine equation parameters for 1-propanol and 1-butanol
        A = [16.1154, 15.3144]
        B = [3483.67, 3212.43]
        C = [205.807, 182.739]

        # Mole fractions in the liquid phase
        x = [0.33, 0.67]

        # Total pressure in atm
        P = 0.5

        # Antoine equation
        def antoine_eqn(T, A, B, C):
            return np.exp(A - B / (T + C))

        # System of equations to solve
        def equations(vars, A, B, C, x, P):
            T, y1, y2 = vars
            P1_sat = antoine_eqn(T, A[0], B[0], C[0])
            P2_sat = antoine_eqn(T, A[1], B[1], C[1])
            return (y1*P - x[0]*P1_sat, y2*P - x[1]*P2_sat, y1 + y2 - 1)

        # Initial guess
        T_guess = 80
        y_guess = [0.5, 0.5]

        # Solve the system of equations
        T, y1, y2 = fsolve(equations, (T_guess, y_guess[0], y_guess[1]), args=(A, B, C, x,

        print(f'Temperature: {T} °C')
        print(f'Composition of the vapor phase: [{y1}, {y2}]')

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Temperature: 10.1437689040661 °C

Composition of the vapor phase: [0.6492781453138041, 0.350721854686196]

Raoult's Law: $P_i = x_i \cdot P_i^*$ where: P_i = partial pressure of component i x_i = mole fraction of component i in the liquid phase P_i^* = vapor pressure of pure component i

Antoine Equation (natural log form): $\ln(P_i^*) = A_i - B_i / (T + C_i)$ where: P_i^* = vapor pressure (kPa) T = temperature (°C) A_i , B_i , C_i = Antoine coefficients

Key Values: $P_{\text{total}} = 0.5 \text{ atm} = 50.66 \text{ kPa}$ $x_{\text{prop}} = 0.33$ (mole fraction of 1-propanol) $x_{\text{but}} = 0.67$ (mole fraction of 1-butanol)

Antoine Coefficients: 1-propanol: $A = 16.1154$, $B = 3483.67$, $C = 205.807$ 1-butanol: $A = 15.3144$, $B = 3212.43$, $C = 182.739$